LAB2: Two medical alarms and the radio's tuner

Week 5 · 1 h 50 min, teams of three
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Goal

Part A uses the 2N3904 of lecture L04 in two medical alarms. In an air-in-line alarm, which stops an infusion pump when a bubble reaches its line (here you will just activate an alarm), it is a switch. In the breathing sensor of an apnoea alarm, which sounds when a sleeping patient stops breathing, it is a temperature-controlled current source. Part B is milestone 1 (M1) of the medium-wave radio your team builds this term: a coil and a tuning capacitor pick one frequency, and lecture L02's peak rectifier, used as a detector, follows the signal's peaks. Predict each result by hand, and in Qucs-S where asked, before you measure it.

Before you come

Pre-lab

Each team prints the report sheet, `LAB2-report.pdf`, and brings to the lab its shaded rows, like this, in Tables 1 and 2, filled in by hand and in Qucs-S as Part A's Predict steps ask, with a screenshot of the Qucs-S sweep on your laptop. Your grader checks them at the start of the session.

Submitting

- **Part A** must be finished in the session: your grader signs off Checkpoints 1 and 2 before it ends, and accepts nothing of Part A later.
- **Part B**, M1, begins once Part A is signed off and is submitted after the lab: your M1 grader signs it off by the start of your group's LAB3, on 4 November, at the latest. To submit, meet your M1 grader in their announced office hours or at another time you agree.
- **Your M1 grader:** Dr. Asem Alaa for Group A, Sara Habo for Group B and Mikhail Nady Daoud for Group C.

The course page's guide shows how to install Qucs-S and ngspice: asem.alaa.page/ed/2026/eleg103/qucs-s. The 2N3904 is in the Libraries tab, under System Libraries, then Transistors.

Equipment and parts

On the bench
Power supply, set to 5 V
Oscilloscope and a ×10 probe
Digital multimeter
Breadboard and jumper wires

Your own kit	Qty
2N3904 transistor	2
Infrared (IR) light-emitting diode (LED), 940 nm, and a phototransistor	1 each
5 V active buzzer, or LED and 330 Ω	1
Red LED; negative-temperature-coefficient (NTC) thermistor, 10 kΩ	1 each
220 Ω	2
330 Ω, 1.8 kΩ, 10 kΩ, 47 kΩ	1 each
Clear tube and syringe	1

Part A1: An air-in-line alarm

In Figure 1 an IR LED shines across a clear tube onto Q_1, a phototransistor: lecture L03's photodiode built into a transistor, so that light makes its base current. Water in the tube focuses the beam like a lens, and a bubble spreads it, so the light on Q_1 falls as a bubble passes. With water, Q_1 conducts and holds its collector low, so Q_2 stays off. With air, Q_1 turns off: current flows through the 47 kΩ and the 10 kΩ into Q_2's base, Q_2 switches on, and the buzzer sounds. These are two switches in cascade, as in L04.

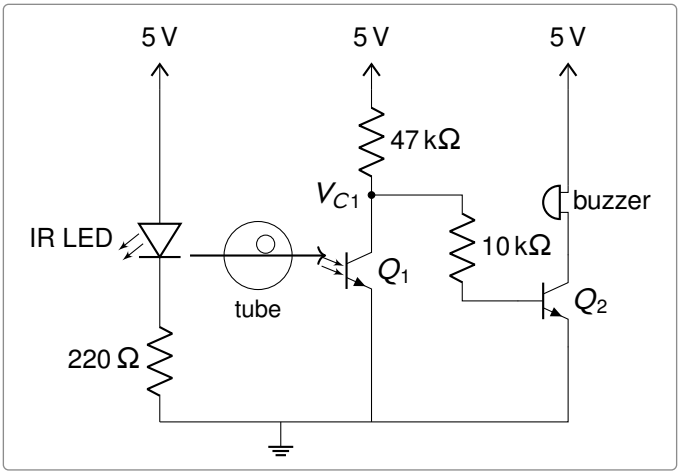


Figure 1: The air-in-line alarm. Q_1 is the phototransistor, Q_2 a 2N3904, and V_{C1} is Q_1's collector voltage.

Predict (pre-lab)

1. **With water**, Q_1 is saturated: what is its collector voltage V_{C1}?
2. **With air**, Q_1 is off. With Q_2's base-emitter voltage $V_{BE} = 0.7\,\text{V}$, find Q_2's base current I_{B2} and V_{C1}.
3. With $\beta = 100$, find the largest buzzer current that keeps Q_2 saturated, and its collector-emitter voltage V_{CE2} while it is. Fill the Hand row of Table 1.

Build and measure

1. Build Figure 1 with the supply's output off. The IR LED and Q_1 face each other across the tube, touching it, at the same height. The buzzer's + lead goes to 5 V. A 2N3904 seen from its flat face, leads down, has its emitter (E), base (B) and collector (C) from left to right.
2. Switch 5 V on. The IR LED's light is invisible to the eye; a phone's front camera shows it glowing. Fill the tube with water from the syringe: the buzzer stays silent. Measure V_{C1}.
3. Push the water out with air: the buzzer sounds. Measure V_{C1}; I_{B2}, from the voltage across the 10 kΩ; and V_{CE2}. Then put the multimeter, on its mA range, in series with the buzzer and measure its current. Fill the Measured row of Table 1.
4. Fill the syringe with water and one small air bubble, and push the bubble slowly past the beam: does the alarm catch it? If room light keeps Q_1 on, shade the beam with your hand or a paper tube.

Table 1: The air-in-line alarm. In the Hand row, I_{C2} is the largest buzzer current that keeps Q_2 saturated, with $\beta = 100$.

	Water	Air			
	V_{C1} (V)	V_{C1} (V)	I_{B2} (µA)	I_{C2} (mA)	V_{CE2} (V)
Hand					
Measured					

Checkpoint 1

Show your grader the alarm catching a bubble, and Table 1.

Part A2: A breathing monitor

In Figure 2 an NTC thermistor is the top of the base divider, the divider that sets the base voltage. Its resistance falls as it warms,

$$R_{NTC} = R_{25}\, e^{B\,(1/T - 1/T_{25})},$$

with $R_{25} = 10\,\text{k}\Omega$ at $T_{25} = 298.15\,\text{K}$ (25 °C), $B = 3950\,\text{K}$ and T in kelvin: about 4.4 % less per degree near 25 °C. Each breath out warms it, so the base voltage rises, and with it the LED's current, which the emitter resistor sets as in lecture L04's DC analysis. Each breath in cools it again.

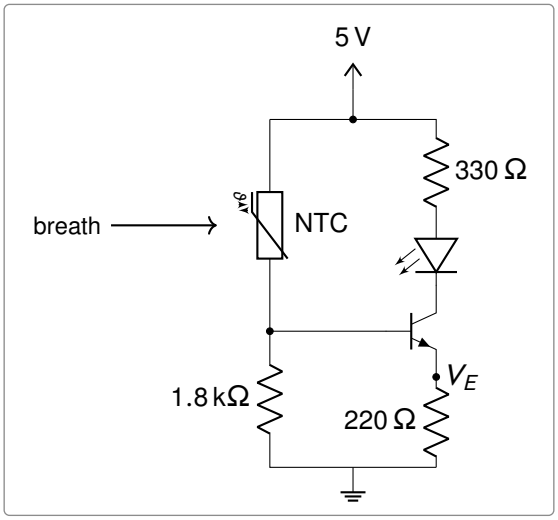


Figure 2: The breathing monitor: a 2N3904, its LED's current set by the thermistor's temperature; V_E is its emitter voltage.

Predict (pre-lab)

1. **By hand,** with $V_{BE} = 0.7\,\text{V}$ and $\beta = 100$, find the thermistor's resistance at 34 °C, about the temperature of breath; then the LED's current I_C at 25 °C and at 34 °C, from the base divider's Thevenin equivalent and the emitter resistor. Fill the Hand row of Table 2.
2. **In Qucs-S,** draw the base divider, with the thermistor as a 10 kΩ resistor named `Rntc`; the 2N3904; and the 220 Ω. Leave out the LED and its 330 Ω, which hardly change the collector current while the transistor is active: connect the collector to 5 V through a current probe, Pr1, instead.
3. Add a **dc simulation** and a **Parameter sweep** whose parameter is `Rntc`, the resistor's name, from 5k to 12k in 141 points, and plot `ngspice/sw1.i(pr1)`. Read I_C at 10 kΩ and at your resistance for 34 °C, and fill the Qucs-S row.

Build and measure

1. Measure the thermistor's resistance out of the circuit at room temperature, then after holding it between two fingers for 30 s.
2. Build Figure 2. Measure V_E across the 220 Ω, which gives $I_C \approx V_E/220\,\Omega$, at room temperature, then after holding the thermistor between two fingers for 30 s. Fill the Measured row of Table 2.
3. Put the scope's channel 1 (CH1) across the 220 Ω, with the probe and CH1 both on ×10: DC coupling, 50 mV/div vertically and 5 s/div horizontally. V_E sits a few tenths of a volt up, so move CH1's zero to the bottom of the screen. With your nose about a centimetre from the thermistor, without touching it, breathe

normally for 30 s: six divisions. Count the breaths, double the count for breaths per minute, and read V_E's swing. Fill Table 3.

Table 2: The breathing monitor's current. On the bench, I_C is $V_E/220\ \Omega$, and R_{NTC} is measured out of the circuit.

	Cool: 25 °C, or the room		Warm: 34 °C, or a finger	
	R_{NTC} (kΩ)	I_C (mA)	R_{NTC} (kΩ)	I_C (mA)
Hand	10	0.264	6.782	1.476
Qucs-S	10			
Measured				

Table 3: Breathing on the thermistor, read off the scope over 30 s.

Breaths per minute	V_E's swing (mV)	V_E's lowest value (V)

Checkpoint 2

Show your grader the trace following your breathing, and Table 2. Part A ends here.

Part B: The radio's tuner and detector (M1)

A coil of inductance L and a tuning capacitor of capacitance C form a tank circuit, which resonates at one frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}:$$

a signal at f_0 builds the largest voltage across the tank, at test point 1 (TP1) in Figure 3. The detector, L02's peak rectifier with time constant $RC = 50\ \mu s$, follows the signal's peaks, so its output at test point 3 (TP3; test point 2 comes with the next milestone) is a DC level for a steady sine, and the sound for a station's amplitude-modulated (AM) signal. A source in M1's band plays the station, with a steady sine: it drives a test loop of five turns of wire beside your coil. Part B has no pre-lab: it begins once your grader has signed off Part A.

Document M1 as the first section of your team's radio guide: a do-it-yourself (DIY) guide on a static website, with one section per milestone, telling the journey and the challenges you met. The guide's requirements and its grading follow this week.

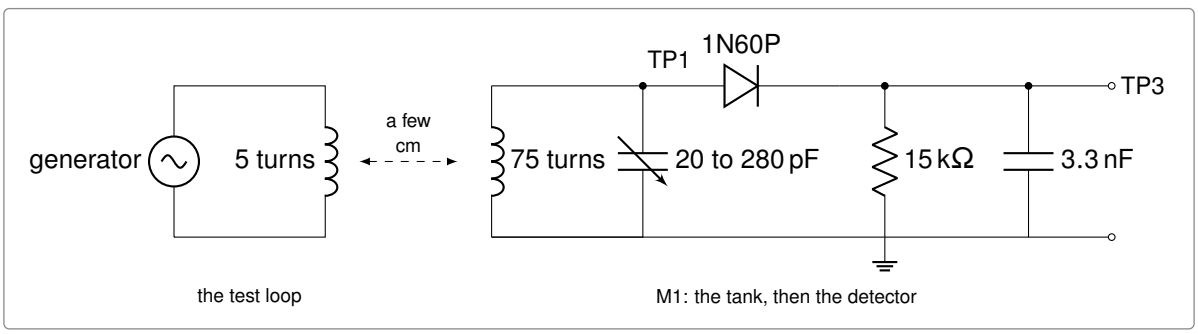


Figure 3: M1 under test: the generator drives the test loop, a few centimetres from M1's coil.

Predict and build

Your team's radio kit holds about 12 m of 0.3 mm enamelled copper wire, a piece of 50 mm polyvinyl chloride (PVC) pipe, a tuning capacitor, a 1N60P diode, a 15 kΩ resistor and a 3.3 nF capacitor.

1. **Predict.** Wheeler's formula gives a single-layer coil's inductance in microhenries,

$$L \approx \frac{d^2 n^2}{457\,d + 1016\,l},$$

   with n its turns, and d its diameter and l its length in millimetres. Your 75 turns, close-wound on the 50 mm pipe, are about 25 mm long. Find L; the capacitance that tunes the tank to 819 kHz; the frequencies at the ends of the capacitor's range, 280 pF and 20 pF; and the detector's RC. Fill Table 4.

2. **Wind the coil.** Tape the wire's start to the pipe, wind 75 turns side by side with no gaps or crossings, and tape the end. Scrape 1 cm of enamel off each end with sandpaper or a blade, until the copper shines.

3. **Fix the capacitor** to the pipe with tape or glue, and join the coil's ends to the two terminals of its AM section; ask your grader which two.

4. **Wire the detector** on your breadboard: the 1N60P from TP1, the top of the tank, to TP3, its band towards TP3; then the 15 kΩ and the 3.3 nF, side by side, from TP3 to ground, the tank's other end.

Measure

Any source that reaches M1's band will do; the lab's function generator gives at most 1 V. Your laptop's sound card can also serve as a function generator and an oscilloscope; the course page's guide shows how: asem.alaa.page/ed/2026/eleg103/home-oscilloscope-fgen. Its input and output cover about 20 Hz to 20 kHz, with no DC, while M1's band runs from about 560 kHz to 2.1 MHz: even the band's bottom is some 28 times the card's top. The card is still a great tool for proofs of concept and for a range of tests at audio frequencies. Would an Arduino Nano or Uno, or a Raspberry Pi, overcome the sound card's frequency limit? Find out for yourselves: you can book any of them from the department at reserve.sbme-labs.app.

Drive the test loop with your source, a few centimetres from your coil and on the same axis. Read TP1 on a scope, with the probe compensated on the scope's 1 kHz output and both the probe and the channel on ×10: a probe on ×1 adds about 100 pF and detunes the tank. Without a scope, TP3's DC on a multimeter follows TP1's peak once that peak is above the detector's dead zone. If your source sends a tone, your laptop's sound card can show it at TP3.

1. **Resonance.** Set the capacitor to half its travel and find the frequency at which TP1, or TP3, peaks: f_0. Then turn the capacitor to each end of its travel and find the peak again: the band's two ends.

2. **Bandwidth.** Back at half travel and f_0, find the frequencies below and above f_0 where TP1's amplitude, or roughly TP3's DC, falls to 0.7 of its peak: f_{low} and f_{high}. The tank's quality factor is $Q = f_0/(f_{high} - f_{low})$. Fill Table 5.

3. **The detector.** At f_0, record TP1's peak and TP3's DC at six settings of your source, from the weakest that moves TP3 to the strongest; with the lab's generator, at 0.05, 0.1, 0.2, 0.3, 0.5 and 1 V. Fill Table 6.

Table 4: M1's predictions: the coil of 75 turns, the tuning capacitor from 20 pF to 280 pF, and the detector.

	L (µH)	C for 819 kHz (pF)	f at 280 pF (kHz)	f at 20 pF (MHz)	RC (µs)
Prediction	249.43	151.4	602.24	2.253	49.5

Table 5: The tank at TP1, driven by the test loop.

Band's low end (kHz)	Band's high end (MHz)	f_0 at half travel (kHz)	f_{low} (kHz)	f_{high} (kHz)	Q
965.5	1.119	1065	956.3	1194	4.48

Table 6: The detector at f_0: TP1's peak on a scope, TP3's DC on a multimeter, at six settings of your source, such as the lab generator's amplitude in volts.

Source setting (V)	TP1's peak (mV)	TP3, DC (mV)
1.035	1348	851.9
0.5	656.3	430.1
0.304	382.8	257.8
0.2031	250	158.2
0.09961	109.4	60.55
0.05	44.53	11.72

Checkpoint 3: M1 signed off

By the start of your group's LAB3 at the latest, show your M1 grader the resonance at TP1, the detector's DC at TP3, and Tables 5 and 6. Keep M1: the next milestones build on it.

Supplementary reading: M1 in context. No step of the lab depends on it.

Where M1 sits in a radio. Every amplitude-modulation receiver begins with M1's two halves. The tuned circuit picks one station out of all those reaching the coil, and the detector takes the sound back out of its carrier. A crystal radio is nothing more: a tuned circuit, a detector and a sensitive earphone, powered by the station itself. The radios of the 1930s, like the one your team builds, add gain before the detector, so that weak stations rise above the diode's dead zone, and gain after it, so that the sound can drive a speaker.

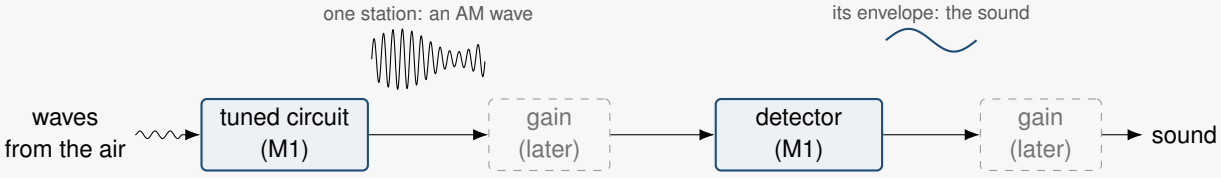


Figure 4: M1's two halves in a radio: the tuned circuit passes one station's wave, and the detector keeps its envelope. Later milestones add the gain.

Who made it. In 1874 Ferdinand Braun found that a metal point pressed on a crystal of a metal sulphide conducts better one way than the other: the first rectifier. Jagadish Chandra Bose, a physics professor in Calcutta, detected millimetre waves with contacts on galena, lead sulphide, and received a United States patent for his detector in 1904 (Figure 5), an early semiconductor device, decades before the transistor. Greenleaf Whittier Pickard patented the silicon *cat's whisker* detector in 1906, and it filled the crystal sets of the 1920s. Tuning came from Oliver Lodge, whose *syntonic* circuits of 1897 let a receiver choose one transmitter, and from Guglielmo Marconi's tuned-circuit patent of 1900. Reginald Fessenden sent the first speech by amplitude modulation in 1900. In Egypt, Egyptian Radio opened on medium wave on 31 May 1934 with the words "هنا القاهرة".

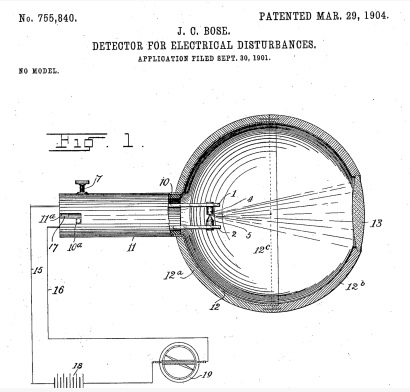
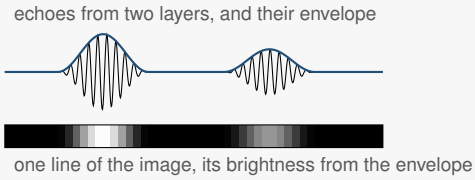
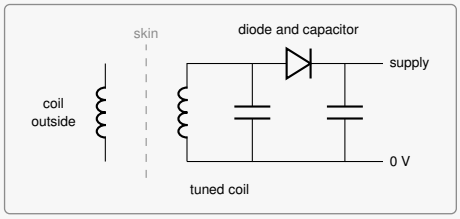


Figure 5: Bose's *Detector for electrical disturbances*, United States patent 755,840 (1904): the galena contacts (4, 5) at the focus of a horn, with a battery (18) and a galvanometer (19).

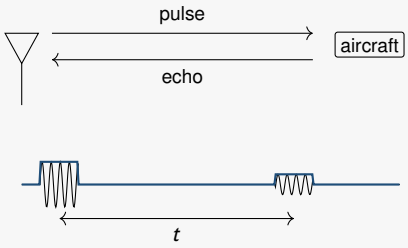
Beyond radio. The same two circuits turn up far from broadcasting (Figure 6).



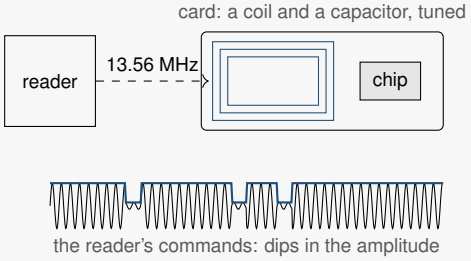
(a) Medicine: ultrasound. Each echo returns as a burst at a few megahertz. The scanner keeps its envelope, as M1's detector does, and turns its height into brightness along one line of the image.



(b) Medicine: cochlear implant. Lecture L03's Part 4 powered an implant through the skin: a coil outside drives a tuned coil inside, and a diode and a capacitor turn the carrier into the implant's supply, as M1's detector turns it into a DC level.



(c) Defence: radar. A pulse goes out and its echo returns after a time t: a detector turns each burst into a pulse, and the range is $ct/2$, with c the speed of light. Second World War radar used silicon cat's-whisker detectors, and the research into them led to the transistor in 1947.



(d) Daily life: bank card. The card's coil and capacitor form a tank tuned to the reader's 13.56 MHz, like M1's. A diode and a capacitor power its chip from that carrier, and the reader sends its commands as short dips in the amplitude, which an envelope detector reads.

Figure 6: M1's two circuits beyond radio.

Report

Each team submits one report, the report sheet `LAB2-report.pdf`, to LAB2's assignment on Google Classroom. It holds Tables 1 to 3 and space for every answer; M1 goes into your radio guide instead. Give each approximation error as a percentage, $100 \times (\text{prediction} - \text{measured})/\text{measured}$.

1. **The alarm.** Give the approximation errors of your hand V_{C1} and I_{B2} with air. Find the forced $\beta = I_{C2}/I_{B2}$ and compare it with $\beta = 100$: was Q_2 saturated? Does your V_{CE2} agree?
2. **The breathing monitor.** (a) From the thermistor's resistance at room temperature, find the room's temperature. (b) Predict I_C again with that resistance, by hand and in Qucs-S, and give each prediction's approximation error. (c) With $V_{BE} = 0.7\,\text{V}$, the hand value misses most at the lower temperature. Why? Compare the voltage left across the $220\,\Omega$ at the two temperatures. (d) Give your breathing rate, in breaths per minute, and V_E's swing.